

Priyadarshini College of Engineering, Nagpur  
Sessional Examination (2023-24) Even Semester  
**B. Tech. Eighth Semester (Computer Technology) (CBCS)**  
Data Analytics using Python

P. Pages: 2

Time: Three Hours

PCE/KW/23/BTCT505T

Max. Marks:70

**Notes:**

- 1) All questions carry marks as indicated.
- 2) Solve Question 1 or Question 2
- 3) Solve Question 3 or Question 4
- 4) Solve Question 5 or Question 6
- 5) Solve Question 7 or Question 8
- 6) Solve Question 9 or Question 10
- 7) Due credit will be given to neatness and adequate dimensions.
- 8) Assume suitable data wherever necessary.
- 9) Illustrate your answers whenever necessary with the help of neat sketches.

| Q. No. | Questions                                                                                                                                                                                                                                                                     | CO  | BL          | Marks       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|-------------|
| 1      | a)List and explain Measures of Central Tendency.<br>b)Explain types of Data Analytics.                                                                                                                                                                                        | CO1 | 2<br>2      | 7<br>7      |
| OR     |                                                                                                                                                                                                                                                                               |     |             |             |
| 2      | a)List and explain various non-random Sampling Techniques.<br>b) Define the following:<br>1)Sample Space<br>2)Unions and Intersections<br>3)Mutually Exclusive Events<br>4)Independent Events<br>5) Collectively Exhaustive Events<br>6) Complementary Events<br>7)Experiment | CO1 | 2<br>1      | 7<br>7      |
| 3      | a)What is regression analysis?Give Simple Linear Regression Model.<br>b) What do you mean by one-way ANOVA?<br>c)Explain the principle behind the least square method.                                                                                                        | CO2 | 1<br>1<br>2 | 5<br>4<br>5 |
| OR     |                                                                                                                                                                                                                                                                               |     |             |             |
| 4      | a)Explain the three Approaches for Hypothesis Testing.<br>b)What is Hypothesis Testing? Define Null hypothesis and Alternate hypothesis<br>c) Differentiate between type I and type II error.                                                                                 | CO2 | 2<br>1<br>2 | 6<br>4<br>4 |
| 5      | a)Define the terms:<br>i)Confidence interval,<br>ii)Prediction interval<br>iii)Point Estimate                                                                                                                                                                                 | CO3 | 1           | 7           |

|    |                                                                                                       |     |   |   |
|----|-------------------------------------------------------------------------------------------------------|-----|---|---|
|    | b)What is residual analysis ? Explain different types of residual analysis                            |     | 2 | 7 |
|    | OR                                                                                                    |     |   |   |
| 6  | a)Explain the Normality of Residual (Error) in both linear and logistic regression.                   | CO3 | 2 | 7 |
|    | b)What do you mean by logistic regression and explain when it is used.                                |     | 2 | 7 |
| 7  | a)Describe various Accuracy Measures for Classification.                                              | CO4 | 2 | 7 |
|    | b)Explain in detail the effect -Interaction of the independent variable to the regression model.      |     | 2 | 7 |
|    | OR                                                                                                    |     |   |   |
| 8  | a)Describe the Chi-square test of independence.                                                       | CO4 | 2 | 7 |
|    | b)Describe the metrics for computation of distances between the objects.                              |     | 2 | 7 |
| 9  | a)What do you mean by categorical variable ? how to find dissimilarity between categorical variables? | CO5 | 2 | 7 |
|    | b)Compare agglomerative and divisive hierarchical clustering methods.                                 |     | 2 | 7 |
|    | OR                                                                                                    |     |   |   |
| 10 | a)What are different attribute selection measures in decision tree?                                   | CO5 | 2 | 7 |
|    | b)Define CART.                                                                                        |     | 1 | 3 |
|    | c)How does tree pruning work in decision tree?                                                        |     | 2 | 4 |